



Profile

AI engineer and medical imaging researcher with publications in medical image synthesis, self-supervised medical representation learning, and anatomy-aware vision-language models. My interests center on medical image analysis, diffusion-based generation, vision-language reasoning, and reliable data/model pipelines for applied AI research.

Education

Sangmyung University

Mar. 2023 - Aug. 2026

Bachelor's Program in Computer Science / Software (GPA: 3.73/4.5)

- Built a foundation in machine learning, deep learning, data preprocessing, algorithms, and software engineering.
- Applied data-driven modeling methods through preprocessing, feature engineering, validation, and model evaluation workflows.
- Developed programming and system design skills for building reliable AI and data-intensive applications.

Publications

3D-LLDM: Label-Guided 3D Latent Diffusion Model for Improving High-Resolution Synthetic MR Imaging in Hepatic Structure Segmentation

K. Kim, J. Bae, Y. Han, J. Y. Bae, S. Ju, **J. Lim**, G. Kim, N. J. Kim, W. K. Jeong, et al.

IEEE International Symposium on Biomedical Imaging (ISBI), 1-5 | 2026

MATHENA: Mamba-based Architectural Tooth Hierarchical Estimator and Holistic Evaluation Network for Anatomy

K. Kim, J. Park, Y. Han, A. Jung, S. Park, S. Lee, J. Yang, J. Han, S. Lee, **J. Lim**, et al.

arXiv preprint arXiv:2604.00537 | 2026

MAESIL: Masked Autoencoder for Enhanced Self-supervised Medical Image Learning

K. Kim, H. Jung, Y. Han, **J. Lim**, Y. Jean, S. Park, E. Choi, H. Go, S. Ju, S. Park, et al.

International Conference on Electronics, Information, and Communication | 2026

NEMESIS: Noise-suppressed Efficient MAE with Enhanced Superpatch Integration Strategy

K. Kim, H. Jung, Y. Han, H. Go, E. Choi, S. Park, **J. Lim**, J. Yang, S. Lee, et al.

arXiv preprint arXiv:2604.01612 | 2026

MINTI: Fine Boundary Instructional Learning for Intelligent Anatomical Vision-Language Models

H. Lee, Y. Han, K. Kim, **J. Lim**, I. Um, H. Go, E. Choi, K. Kim, S. Park, J. Jeong, et al.

IEIE Annual Conference, 950-953 | 2026

Research Experience

Medical Imaging Lab, Seoul National University

Sep. 2025 - Present

Research Assistant

- Advisor: Prof. Nam-Joon Kim (SNU) and Prof. Wonjae Lee, Radiologist (Samsung Medical Center).
- Participate as a member of the SUDAL Medical Imaging Team, contributing to journal clubs, paper reviews, and collaborative research discussions.
- Built the official IMSI Lab website from the ground up to organize lab research, members, and publication records.
- Conduct research on conditional abdominal MRI synthesis using segmentation-guided ControlNet pipelines.
- Collaborate with Samsung Medical Center radiologists and NVIDIA researchers on clinically motivated medical AI projects.

Research Collaborations

SUDAL Medical Imaging Team

NVIDIA Collaboration

Medical Imaging AI Research

- Collaborate with Prof. Nam-Joon Kim and medical AI researchers on clinically oriented medical imaging projects.
- Work with NVIDIA researcher Ping Shu Ho (Cliff) on diffusion-based medical imaging research topics.
- Focus on medical image synthesis, reconstruction, segmentation-aware generation, and robust visual representation learning.

Key Research Projects

3D Latent Diffusion for MR Imaging

2026

Label-guided synthetic MRI and hepatic structure segmentation

- Contributed to the experimental workflow for label-guided 3D latent diffusion models for high-resolution synthetic MR imaging.
- Supported segmentation-aware generation experiments aimed at improving hepatic structure segmentation under limited medical imaging data.
- Reviewed model outputs and failure cases to connect synthesis quality with downstream segmentation utility.

Medical Foundation Models with MAE and Mamba

2026

Self-supervised representation learning and anatomical structure estimation

- Worked on MAE-based medical image representation learning pipelines for robust downstream evaluation.
- Explored Mamba-based modeling approaches for hierarchical anatomical estimation and efficient visual sequence modeling.
- Participated in literature review, experiment organization, and result analysis for medical foundation model research.

Medical Vision-Language Model Development

2026

Instruction tuning for anatomy-aware VLMs

- Developed research workflows for fine-grained boundary instruction learning in anatomical vision-language models.
- Connected medical imaging tasks with multimodal reasoning, report-aligned supervision, and clinical visual understanding.
- Contributed to dataset/task formulation for evaluating anatomy-aware vision-language reasoning.

Professional Experience

INTWEEN

Jun. 2026 - Present

AI SaaS / Data Systems Engineer

- Develop in-house AI-based SaaS products, including Sasure, by integrating internal datasets and service workflows.
- Design pipelines for collecting, cleaning, normalizing, and integrating fragmented operational data into analysis-ready formats.
- Build workflow automation systems that connect data preprocessing, retrieval, and AI-assisted decision support.
- Translate heterogeneous operational data into structured resources for enterprise AI transformation and internal model-driven services.

GDSK

Jan. 2025 - Aug. 2025

Data / AI Engineer

- Designed ETL processes that parse unstructured address text and convert it into standardized address formats.
- Built database architecture and preprocessing workflows for high-volume geographic and address-based datasets.
- Integrated public and geospatial datasets to construct feature sets for regional income estimation models.
- Developed data collection and preprocessing pipelines for large-scale geospatial information analysis.

OUTTA

Jul. 2024 - Present

Medical Imaging Team Lead

- Contribute to an AI organization connected to Seoul National University research and education communities.
- Design bootcamp curriculum and hands-on learning materials for Python, deep learning, and medical AI topics.
- Lead structured paper study sessions on medical image analysis, diffusion models, MAE-based representation learning, and VLMs.
- Coordinate research discussions and project execution across junior members to strengthen reproducible AI research practice.

Certifications & Licenses

TOEIC

935

Oct. 2022

Big Data Analysis Engineer

Certification

Acquired

SQLD

SQL Developer

Acquired

AWS SAA

Solutions Architect Associate

Acquired

NCA

Naver Cloud Platform

Acquired

Information Processing Engineer

National Technical Qualification

Acquired

Technical Skills

Programming Languages: Python, SQL, JavaScript, TypeScript, Java, Bash

AI / ML Frameworks: PyTorch, MONAI, Hugging Face Transformers, scikit-learn, TensorFlow

Medical AI: Medical Image Analysis, 3D Latent Diffusion, Masked Autoencoders, Mamba, Vision-Language Models, Segmentation-aware Generation

LLM / LLMOps: LLM Fine-tuning, RAG, Prompt Engineering, Model Evaluation, Workflow Automation

Data Engineering: ETL Pipelines, Data Cleaning, Geospatial Data Processing, Feature Engineering, Database Architecture

Backend & Infrastructure: FastAPI, Node.js, REST APIs, Docker, Linux, AWS, Naver Cloud Platform

Tools: Git, GitHub, VS Code, PyCharm, Docker Compose, Notion, Figma

Contact Information